

MATTHEW LEE

EECS student with proven hands-on experience designing **PCBs**, developing **embedded firmware**, and leading hardware projects. Interested in **hardware design & validation** applying skills in **PCBA design, bring-up, and system-level debugging**.

CONTACT

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Portfolio: <https://mattyl37.github.io/projects.html>

EDUCATION

University of California Berkeley

Berkeley, CA

Bachelor of Science in Electrical Engineering and Computer Sciences (EECS)

Expected May 2027

- **Relevant Coursework:** Computer Architecture; Signals and Systems; Microelectronic Devices and Circuits
- **Accolades:** Member of Eta Kappa Nu (EECS Honor Society) and IEEE
- **Positions:** Supernode Makerspace Electronics Admin
- **GPA:** 4.0/4.0

EXPERIENCE

Electrical Team Project Manager

Apr. 2025 – Present

UC Berkeley Solar Vehicle Team (CalSol)

Berkeley, CA

- **Gen11 Solar Vehicle**
 - * Developed the solar vehicle's **high-voltage (120V Li-ion)** and **low-voltage CAN power system architecture**, integrating a **BMS** and **distribution boards** for safe, efficient, and reliable operation.
 - * Validated **custom PCBs** on the bench through **oscilloscope measurements** and functional load testing, ensuring each subsystem's correct operation before integration.
 - * Performed extensive debugging on-vehicle **harnessing** to ensure clean signals in **CAN, SPI, isoSPI, and I2C lines**.
 - * Deployed **ESP32 firmware** supporting **CAN-based communication** between boards.
- **Leadership & Management:**
 - * Bridges mechanical and electrical designs to increase efficiency and streamline **system integration**.
 - * Managed **30+ member** electrical team across **10+ projects**, driving the vehicle's **two-year build cycle**
 - * Drove adoption of strong **documentation** practices and authored detailed **Confluence** logs, ensuring design decisions and bring-up processes were **transparent, reusable, and team-wide accessible**.

Horn PCB Lead Designer

Oct. 2024 – Apr. 2025

UC Berkeley Solar Vehicle Team (CalSol)

Berkeley, CA

- Designed, brought up, and verified a **mixed-signal PCB**, integrating both **hardware** and **embedded firmware**.
- Completed **PCB schematic, PCB layout, and PCB bring up** of a custom board from scratch, gaining hands-on experience with the full **hardware development cycle**.
- Engineered and flashed a self-made **ESP32 Dev Board** with a **CAN transceiver** and **UART bootloading**, supporting seamless and uniform integration.
- Implemented an **I2C-based DAC, bandpass filtering, and high-gain amplification** to optimize **horn signal integrity** and minimize **power loss**.

Guitar Pedal Engineer

Mar. 2025 – Aug. 2025

Personal Projects

Berkeley, CA

- Engineered custom **analog** guitar pedals using **op-amp circuits** and active **bandpass filters** to shape signal output.
- Applied **frequency analysis** and **LTspice** simulations to validate circuit behavior prior to physical verification.
- Applied **BJT circuit design** to implement **signal clipping** and achieve a custom distortion effect.

TECHNICAL SKILLS

Tools & Platforms: KiCAD, Platform.io, Git, SPICE, VS Code, Python, Jira, Confluence, SolidWorks

Hardware: PCB Design & Layout, Mixed-signal Circuit Design, Power Electronics, Signal Filtering, CMOS

Embedded Systems & Languages: C/C++, RISC-V Assembly, ESP32, CAN, SPI, isoSPI, I2C, I2S, UART, Python

Validation & Testing: PCB bring-up, System-level verification and integration, Documentation, BUS Monitoring, Harnessing, Soldering, Oscilloscope Testing